

消抖动设计

1、实验任务：

掌握消抖动电路的原理。

2、实验原理：

分别用两个计数器去对输入信号的高电平和低电平的持续时间（脉宽）进行计数（在时间上是同时但独立计数）。只有当高电平的计数时间大于某值，则判为遇到正常信号，输出 1；若低电平的计数时间大于某值，则输出 0。此例的仿真波形如图 1 所示。

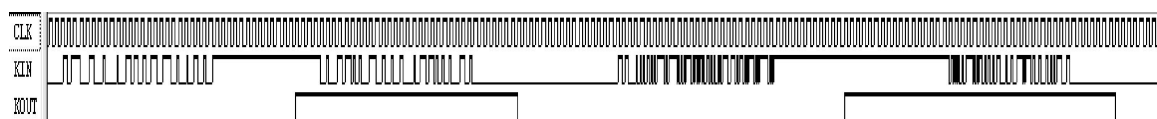


图 1 消抖动电路仿真波形

由波形图可见，其输出信号脉宽比逻辑方式输出的信号要宽得多。此例的输出脉宽与正常信号高电平 KH 的位宽和工作时钟频率共同决定，不单纯由时钟决定。所以优于以上的逻辑方式。此实验给出的设计比前面的电路要容易控制，且效果更好，只是耗用资源比较多。此电路同样能用于消除来自不同情况的干扰、毛刺和电平抖动。其中的工作时钟 CLK 的频率大小要视干扰信号和正常信号的宽度而定。对于类似键抖动产生的干扰信号，频率可以低一些，数万赫即可；若为比较高速的时钟信号，则可利用 FPGA 内的锁相环，使 CLK 能到达 200MHz 以上。此外，KH 和 KL 的计数位宽和计数值都可以具体情况调节。

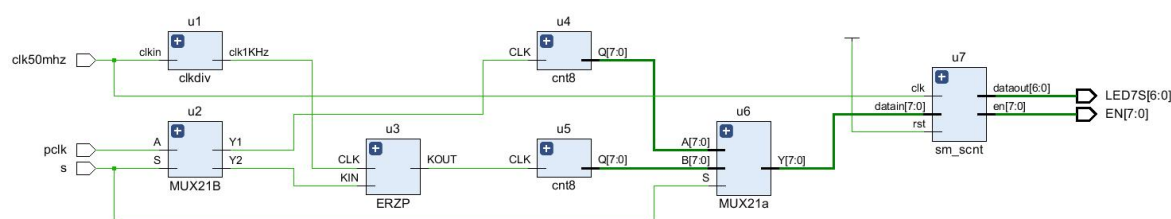


图 2 消抖动 RTL 电路图

3、实验程序：

```
module ERZP (CLK, KIN, KOUT);
    input  CLK, KIN;           //工作时钟和输入信号
    output KOUT;              reg KOUT;
    reg [3:0] KH, KL;         //定义对高电平和低电平脉宽计数之寄存器
    always @(posedge CLK) begin
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if (!KIN) KL<=KL+1; //对键输入的低电平脉宽计数
else KL<=4'b0000; end //若出现高电平，则计数器清零
    always @(posedge CLK) begin
        if (KIN) KH<= KH+1; //同时对键输入的高电平脉宽计数
        else KH<=4'b0000; end //若出现高电平，则计数器清零
    always @(posedge CLK) begin if (KH > 4'b1100) KOUT<=1'B1; //对高电平脉宽计数一旦大于
    12，则输出1 else if (KL > 4'b0111) KOUT<=1'B0; //对低电平脉宽计数若
    大于7，则输出0
    end
endmodule

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5、引脚锁定:

#系统时钟约束

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set_pin_assignment { EN[0] } { LOCATION = D11; IOSTANDARD = LVCMOS33;
DRIVESTRENGTH = 8; PULLTYPE = NONE; }
set_pin_assignment { EN[1] } { LOCATION = E11; IOSTANDARD = LVCMOS33;
DRIVESTRENGTH = 8; PULLTYPE = NONE; }
set_pin_assignment { EN[2] } { LOCATION = F10; IOSTANDARD = LVCMOS33;
DRIVESTRENGTH = 8; PULLTYPE = NONE; }
set_pin_assignment { EN[3] } { LOCATION = C13; IOSTANDARD = LVCMOS33;
DRIVESTRENGTH = 8; PULLTYPE = NONE; }
set_pin_assignment { EN[4] } { LOCATION = E10; IOSTANDARD = LVCMOS33;
DRIVESTRENGTH = 8; PULLTYPE = NONE; }
set_pin_assignment { EN[5] } { LOCATION = E12; IOSTANDARD = LVCMOS33;
DRIVESTRENGTH = 8; PULLTYPE = NONE; }
set_pin_assignment { EN[6] } { LOCATION = E16; IOSTANDARD = LVCMOS33;
DRIVESTRENGTH = 8; PULLTYPE = NONE; }
set_pin_assignment { EN[7] } { LOCATION = F16; IOSTANDARD = LVCMOS33;
DRIVESTRENGTH = 8; PULLTYPE = NONE; }
set_pin_assignment { LED7S[0] } { LOCATION = C15; IOSTANDARD = LVCMOS33;
DRIVESTRENGTH = 8; PULLTYPE = NONE; }
set_pin_assignment { LED7S[1] } { LOCATION = C16; IOSTANDARD = LVCMOS33;
DRIVESTRENGTH = 8; PULLTYPE = NONE; }
set_pin_assignment { LED7S[2] } { LOCATION = B6; IOSTANDARD = LVCMOS33;
DRIVESTRENGTH = 8; PULLTYPE = NONE; }
set_pin_assignment { LED7S[3] } { LOCATION = A5; IOSTANDARD = LVCMOS33;
DRIVESTRENGTH = 8; PULLTYPE = NONE; }
set_pin_assignment { LED7S[4] } { LOCATION = B5; IOSTANDARD = LVCMOS33;

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DRIVESTRENGTH = 8; PULLTYPE = NONE; }
set_pin_assignment { LED7S[5] } { LOCATION = A8; IOSTANDARD = LVCMOS33;
DRIVESTRENGTH = 8; PULLTYPE = NONE; }
set_pin_assignment { LED7S[6] } { LOCATION = A7; IOSTANDARD = LVCMOS33;
DRIVESTRENGTH = 8; PULLTYPE = NONE; }
set_pin_assignment { clk50mhz } { LOCATION = R7; IOSTANDARD = LVCMOS33; PULLTYPE =
PULLUP; }
set_pin_assignment { pclk } { LOCATION = A2; IOSTANDARD = LVCMOS33; PULLTYPE =
PULLUP; }
set_pin_assignment { s } { LOCATION = C5; IOSTANDARD = LVCMOS33; PULLTYPE =
PULLUP; }

```

6、下载测试：

SW4 拨上，按动键 1，数码计数稳定，拨下，按动键 1 计数，数码计数不稳定。

